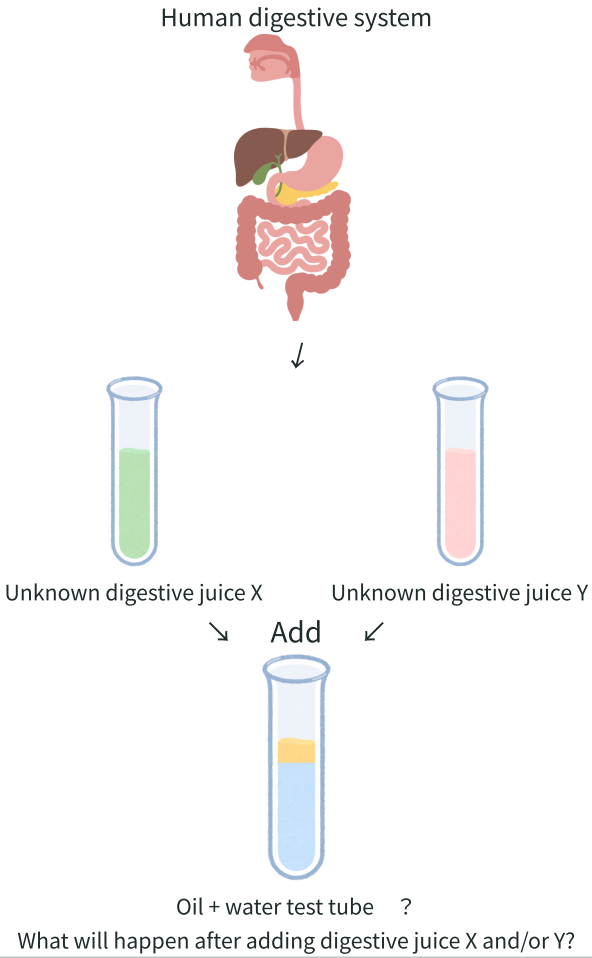
Name: _____ Class and student number: _____

Unknown Digestive Juices X and Y Their effects on lipids



Research background

Lipids in our food have large molecular structures. They cannot pass directly through the cells lining the digestive tract (消化道) to be absorbed.

Research task

How do digestive juices X and Y affect the appearance of an oil–water mixture?

Researchers have collected digestive juices X and Y from the digestive tract (消化道). Design a fair test in this virtual lab to discover their functions.

Your first observation

Describe only what you can see in the oil + water test tube (do not yet guess what X and Y are or what the results will be).

Name: _____ Class and student number: _____

01 • Write an idea to test

Complete your prediction

If I add _____to the oil and water, the mixture in the test tube will _____.

Choose one.

☐ Digestive juice X ☐ Digestive juice X + digestive juice Y ☐ Digestive juice Y

Choose one.

☐ become clear ☐ become cloudy ☐ show no obvious change

My reason (required; there is no single correct answer)

_____

_____

_____

02 • Fair test design

You may choose more than one.

Independent variable (the factor changed on purpose)

☐ Total volume of oil and water
☐ Reaction time
☐ Method of shaking
☐ Combination of digestive juices
☐ Reaction temperature
☐ Total volume of liquid added
☐ Appearance of the mixture

Dependent variable (the result you measure or observe)

☐ Total volume of oil and water
☐ Reaction time
☐ Method of shaking
☐ Combination of digestive juices
☐ Reaction temperature
☐ Total volume of liquid added
☐ Appearance of the mixture

Controlled variables (the factors kept the same)

☐ Total volume of oil and water
☐ Reaction time
☐ Method of shaking
☐ Combination of digestive juices
☐ Reaction temperature
☐ Total volume of liquid added
☐ Appearance of the mixture

Name: _____ Class and student number: _____

03 • Conditions for a fair test

Which conditions does this fair test need?

Choose all the conditions that apply. You may choose more than one.

You may choose more than one.

- ☐ Observe all setups after the same reaction time.
- ☐ The X + Y group can have more liquid in total to compare it with the other groups.
- ☐ Use the same starting amounts of oil and water and the same total volume of added liquid in each setup.
- ☐ Use the same method of shaking for every setup.

04 • Control group

Does this experiment need a control group? If so, how would you design the setups to compare?

(Hint: A control group provides a starting point for comparison. It helps you check that changes are caused by the independent variable, rather than other outside factors. The control group therefore does not receive the treatment being tested.)

The teacher saves this original design before giving out Part B. Keep a copy to refer to during reflection.

Name: _____ Class and student number: _____

05 • Experiment setup design

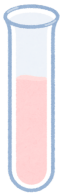
Draw and label your setup

Use your experimental variables to help you draw and label your setup.

(Hint: How many setups do you need? Which groups will you test and which is the control group?)


Oil–water test tubes ×4


Digestive juice X ×1


Digestive juice Y ×1


Droppers ×4


37°C water bath ×1

Written design (can replace a drawing): describe the four setups, how much liquid you add and which conditions you keep the same.

The teacher saves this original design before giving out Part B. Keep a copy to refer to during reflection.

Name: _____ Class and student number: _____

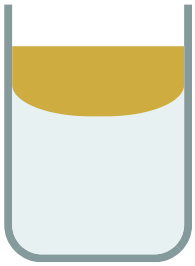
Carry out the investigation and collect data

Change one factor at a time. First set up a group without heating to use for comparison.

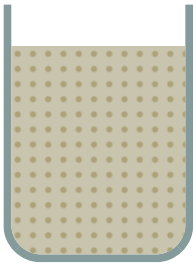
✦Fair test: oil, water, temperature and time are fixed. Only the combination of digestive juices changes.

Observe the static images, choose and record the appearances. The images show the preset results of this VL.

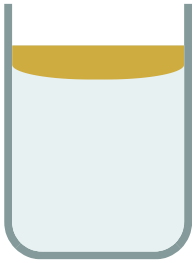
Control group: add 2 mL of water



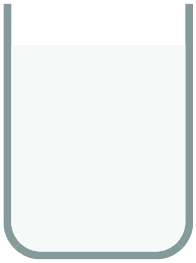
Add 2 mL of digestive juice X



Add 2 mL of digestive juice Y



Add 1 mL each of digestive juices X and Y



Name: _____ Class and student number: _____

Select the appearance based on what you observe:

Experiment setup	First confirmed answer • Your recorded appearance
Control group: add 2 mL of water	☐ Separate oil and water layers (no clear change you can see) ☐ Separate oil and water layers (thinner oil layer) ☐ become cloudy ☐ become clear
Add 2 mL of digestive juice X	☐ Separate oil and water layers (no clear change you can see) ☐ Separate oil and water layers (thinner oil layer) ☐ become cloudy ☐ become clear
Add 2 mL of digestive juice Y	☐ Separate oil and water layers (no clear change you can see) ☐ Separate oil and water layers (thinner oil layer) ☐ become cloudy ☐ become clear
Add 1 mL each of digestive juices X and Y	☐ Separate oil and water layers (no clear change you can see) ☐ Separate oil and water layers (thinner oil layer) ☐ become cloudy ☐ become clear

If you revise an answer, keep the first answer and write the change here.

Name: _____ Class and student number: _____

03 • Data and explanations

Complete the conclusions using your data

1. What does digestive juice X do?

Digestive juice X mainly causes lipids to form _____.

Choose one.

☐ a clear solution ☐ a cloudy emulsion (乳狀液)

2. How does digestive juice X help digestive juice Y act?

Digestive juice X first breaks up large oil droplets into small droplets, causing their total surface area to _____, which helps digestive juice Y act.

Choose one.

☐ decrease ☐ increase

3. What can this experiment tell us?

We only observed the oil layer and whether the mixture was cloudy or clear. We did not test for digestion products. Which statement fits best?_____

Choose one.

☐ Appearance provides clues about the effects of digestive juices, but alone cannot prove that glycerol (甘油) and fatty acids (脂肪酸) have formed.

☐ A clear mixture completely proves that lipids have broken down into glycerol (甘油) and fatty acids (脂肪酸).

If you revise an answer, keep the first answer and write the change here.

Name: _____ Class and student number: _____

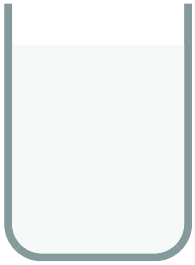
Extension: the effects of heating on digestive juices

Which digestive juice is affected by high temperature?

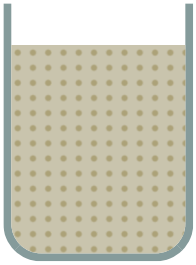
Use “add 1 mL each of digestive juices X and Y” as the group for comparison. Boil only one digestive juice at a time. Complete both tests and record your observations.

Cool each boiled digestive juice to the same reaction temperature of 37°C before comparing.

Add 1 mL each of digestive juices X and Y (boil X, then cool)



Add 1 mL each of digestive juices X and Y (boil Y, then cool)



Select the appearance based on what you observe:

Experiment setup	First confirmed answer • Your recorded appearance
Add 1 mL each of digestive juices X and Y (boil X, then cool)	☐ become cloudy ☐ become clear
Add 1 mL each of digestive juices X and Y (boil Y, then cool)	☐ become cloudy ☐ become clear

Extension conclusion: the digestive juice that loses its function after boiling is_____.

Choose one.

☐ Digestive juice X ☐ Digestive juice Y

If you revise an answer, keep the first answer and write the change here.

Name: _____ Class and student number: _____

Give out this part only after the original investigation answers have been submitted.

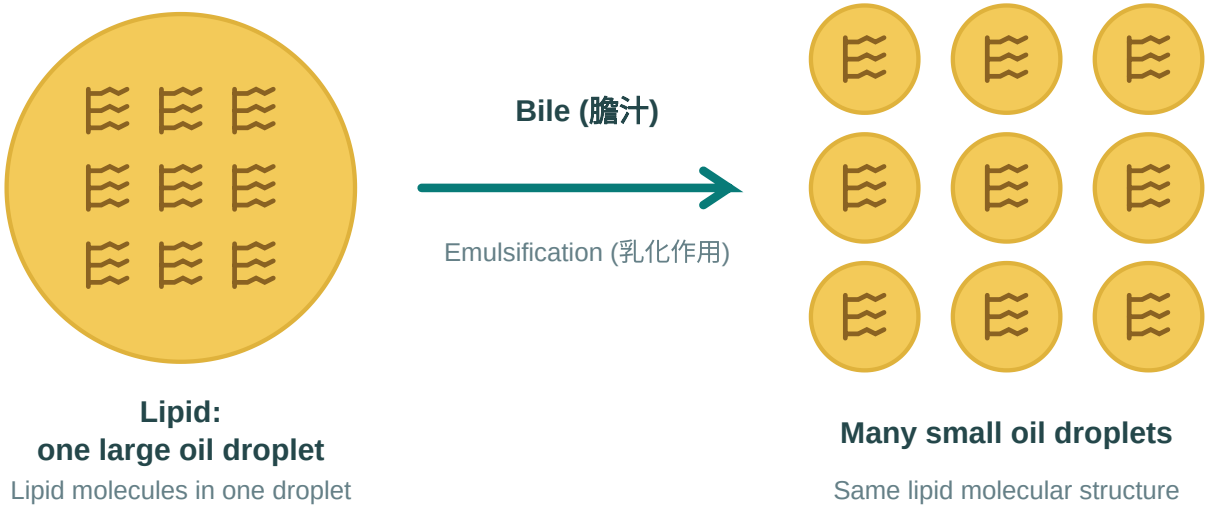
Key learning points

You have now explored the functions of digestive juices X and Y.

1. Digestive juice X is bile (膽汁). It is produced in the liver and stored in the gall bladder (膽囊). Bile (膽汁) contains no digestive enzymes. Through physical digestion (物理消化), it emulsifies lipids (乳化脂質) into small oil droplets, increasing their surface area so chemical digestion (化學消化) can work more efficiently.

2. Digestive juice Y is lipase (脂肪酶). It catalyses the chemical digestion (化學消化) of lipids, breaking them down into smaller molecules of glycerol (甘油) and fatty acids (脂肪酸) to aid absorption. Lipase (脂肪酶) is a protein; high temperature causes denaturation, so it loses its function.

01 • Physical digestion (物理消化): emulsification (乳化作用), with no change in molecular structure



Same lipid amount; greater total surface area

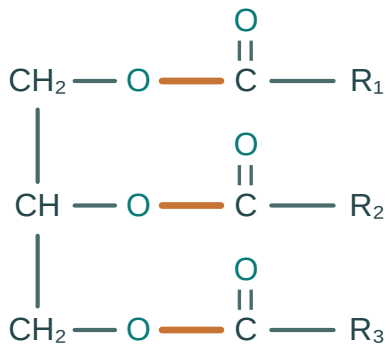
Only the size of the oil droplets and how spread out they are change; lipid molecules are not broken down. Small droplets increase the total surface area. The diagram does not show the real sizes.

Name: _____ Class and student number: _____

Give out this part only after the original investigation answers have been submitted.

02 • Chemical digestion (化學消化): hydrolysis forms different molecules

Lipid: triglyceride (甘油三酯)



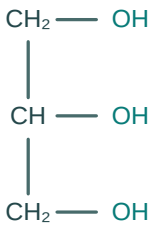
Orange bonds: where hydrolysis happens

+ $3H_2O$ (water)

lipase (脂肪酶)

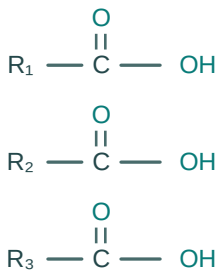
Hydrolysis

glycerol (甘油)



+

3 fatty acids (脂肪酸)



Triglyceride (甘油三酯) + $3H_2O$ ⟶ Glycerol (甘油) + 3 fatty acids (脂肪酸)

R_1, R_2 and R_3 show the hydrocarbon chains of fatty acids (脂肪酸). Hydrolysis changes chemical bonds to form glycerol (甘油) and fatty acids (脂肪酸). Lipase (脂肪酶) is a catalyst and is not used up in the reaction. This diagram uses the simple complete-hydrolysis model commonly taught in this course.

What the evidence can tell us

What the evidence can tell us: this simulation uses set appearances to show emulsification (乳化作用) and enzyme action. A clear mixture does not directly show that glycerol (甘油) or fatty acids (脂肪酸) have formed, or measure digestion speed. Real experiments need other chemical tests and repeated tests. In the heating comparison, we assume that boiled liquids are cooled to the same reaction temperature before use.

Name: _____ Class and student number: _____

Learning reflection

Do your observations support your first idea to test? Name the groups you compared and what you saw. Use the key learning points to improve your explanation.

Refer to your original idea to test and reason saved in Part A. You do not need to rewrite them from memory.

For teachers only. Do not give this part to students.

Distribution and evidence preservation

1. Teachers mark paper objective answers using the same key and weights. Automatic scoring in the shared guide refers to VL Excel; handwritten answers are not read automatically.
2. Use the same questions, concepts, hints and available time for both groups. The control group uses static images and tables instead of interactive simulations; no extra physical experiment is added.
3. Give out A first. Save the original prediction, reason and design before giving out B. Photograph or copy A; let students keep a copy of their original prediction.
4. In B, complete the four basic sets and three conclusions before the heating page and its conclusion.
5. Keep the first confirmed observations readable. Use one line to cross out a revised answer, or write it in the revision area; do not erase it.
6. Save or collect A and B before giving out C. Let students refer to the saved original prediction and write their learning reflection.
7. D and the complete pack are for teachers. Print each part separately using its page range and give it out at the correct stage, rather than giving the whole pack to students at once.
8. Images use oil layers, shade and texture. Before printing in black and white, check that cloudy and clear mixtures and the two oil-layer thicknesses remain distinguishable.
9. Paper tasks do not generate clicks or system time logs. Score the answers and design evidence; do not convert completion, time or clicks into skill scores.
10. Do not deduct marks from a reasonable, testable original prediction just because its result was wrong. Teachers judge written answers and reflection rather than counting keywords.

Keep weekly worksheets separate from the formal 10-item SPSAI and 4-item CKT pre- and post-tests.

Draft activity scoring guide for researcher review; it is not a validated SPSAI or CKT.

A blank teacher score means pending, while an entered 0 means zero. Calculate the overall score only after reflection and required scoring are complete.

For teachers only. Do not give this part to students.

Objective answers and design reference

Independent variable	Combination of digestive juices
Dependent variable	Appearance of the mixture
Controlled variables	Total volume of oil and water / Reaction temperature / Reaction time / Total volume of liquid added / Method of shaking
03 • Conditions for a fair test	Observe all setups after the same reaction time. Use the same starting amounts of oil and water and the same total volume of added liquid in each setup. Use the same method of shaking for every setup.
1. What does digestive juice X do?	a cloudy emulsion (乳狀液)
2. How does digestive juice X help digestive juice Y act?	increase
3. What can this experiment tell us?	Appearance provides clues about the effects of digestive juices, but alone cannot prove that glycerol (甘油) and fatty acids (脂肪酸) have formed.
Extension conclusion: the digestive juice that loses its function after boiling is_____.	Digestive juice Y

Control group: add 2 mL of water	Separate oil and water layers (no obvious change)
Add 2 mL of digestive juice X	become cloudy
Add 2 mL of digestive juice Y	Separate oil and water layers (thinner oil layer)
Add 1 mL each of digestive juices X and Y	become clear
Add 1 mL each of digestive juices X and Y (boil X, then cool)	become clear
Add 1 mL each of digestive juices X and Y (boil Y, then cool)	become cloudy

For teachers only. Do not give this part to students.

Reference for written designs

04 • Control group

Use oil–water + 2 mL of water as the control group. Compare it with oil–water + 2 mL of X, + 2 mL of Y, and + 1 mL each of X and Y. Keep the initial amounts of oil and water, total added volume, reaction temperature, reaction time and shaking method the same.

05 • Experiment setup design

Clearly label the four basic setups: control, X, Y and X + Y. For heat treatment, compare combinations in which X or Y is boiled and then cooled to 37°C. Change one factor at a time.

An oil layer, a cloudy mixture or a clear mixture only gives clues about appearance. This alone cannot prove that glycerol (甘油) and fatty acids (脂肪酸) have formed; other chemical tests are needed. This simulation uses set teaching results and does not measure reaction speed.

A blank teacher score means pending, while an entered 0 means zero. Calculate the overall score only after reflection and required scoring are complete.

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Scoring guide

Draft activity scoring guide for researcher review; it is not a validated SPSAI or CKT.

Item	Maximum	Scoring	Full credit	Partial credit	Zero / missing evidence
Draft activity scoring guide: researcher review needed	32	Research note	Six SPS areas, 4 each; four new-knowledge items, 2 each. Total 32.	Review question difficulty, opportunities and scorer agreement across VLs.	Separate from validated SPSAI / CKT; give paper students the same questions and static data.
Initial observation	2	Teacher	2: accurately describes visible oil/water layers and details, separating observation from guesses.	1: a specific but incomplete description.	0: no useful observation; keep missing older evidence pending rather than inventing it.
Six appearances	2	Answer key	All six answers confirmed; matching appearances / 6 × 2, rounded to two decimal places.	Credit in proportion to the matching appearances.	0 if all answered but none match; a missing set stays unavailable, rather than treating completion as skill.
Independent variable	1	Answer key	Select only the combination of digestive juices.	No partial credit.	0 for an incorrect answer; do not replace missing evidence with 0.
Dependent variable	1	Answer key	Select only the appearance of the mixture.	No partial credit.	0 for an incorrect answer; do not replace missing evidence with 0.

A blank teacher score means pending, while an entered 0 means zero. Calculate the overall score only after reflection and required scoring are complete.

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Scoring guide

Draft activity scoring guide for researcher review; it is not a validated SPSAI or CKT.

Item	Maximum	Scoring	Full credit	Partial credit	Zero / missing evidence
Controlled variables	2	Answer key	Select all five fixed conditions, with no wrong choices.	Correct controlled choices / 5 × 2; a choice from the wrong category gives 0.	Older versions had three choices; do not infer skill from missing new choices. Review cross-version comparisons.
Original testable prediction and reason	2	Teacher	2: conditions can be compared, prediction can be observed, and reason is reasonable.	1: testable but the reason or conditions are incomplete.	0: not testable or no useful content. A reasonable wrong prediction is not penalised. Missing original evidence stays pending.
Conditions for a fair test	1	Answer key	Select time, initial oil/water and total added volume, and shaking method; do not select more total liquid.	No partial credit.	0 for an incorrect set; a version without this question stays unavailable.
Control group	1	Teacher	1: explains its comparison purpose, proposes a setup without digestive juice, and keeps conditions the same.	Enter only 0 or 1.	0: no useful comparison explanation or an unreasonable reason; missing question stays pending.
Setup quality	2	Teacher	2: clear drawing/photo/text, four labelled sets and reasonable volumes.	1: workable but labels or volumes are incomplete.	0: no workable design; do not penalise a written design.

A blank teacher score means pending, while an entered 0 means zero. Calculate the overall score only after reflection and required scoring are complete.

For teachers only. Do not give this part to students.

Scoring guide

Draft activity scoring guide for researcher review; it is not a validated SPSAI or CKT.

Item	Maximum	Scoring	Full credit	Partial credit	Zero / missing evidence
Fixed conditions and experiment arrangement	2	Teacher	2: fair comparison based on design/evidence; boiled liquids are cooled; only one factor changes each time.	1: generally reasonable but fixed conditions or heating arrangements are incomplete.	0: comparisons are not valid. Set completion is evidence only; time or clicks do not directly earn marks.
Conclusions, comparisons and limits	4	Answer key	Current three conclusions and the heating conclusion: 1 each, total 4. Older six-question weights remain 0.5, 0.5, 0.5, 0.75, 0.75, 1.	Award each question its stated answer-key credit.	Use the questions belonging to that version; do not demand deleted questions or invent missing older answers.
Data and claims	4	Teacher	4: organises data clearly, compares at least two sets, links a claim and explains limits.	3: clear comparison but weaker limits; 2: useful comparison but weak links; 1: one relevant description.	0: no useful communication. Use reasons, control explanation, labels, table and reflection. Judge clarity without scoring knowledge twice.
Bile (膽汁), emulsification (乳化作用) and surface area	2	Teacher: post-learning reflection only	2: bile (膽汁) has no digestive enzymes; emulsification (乳化作用) is physical digestion (物理消化), and small droplets increase total surface area.	1: partly correct but incomplete explanation.	0: concept unused or wrong; an unsubmitted reflection does not mean zero.
Lipase (脂肪酶), chemical digestion (化學消化) and products	2	Teacher: post-learning reflection only	2: lipase (脂肪酶) speeds up chemical digestion (化學消化), forming glycerol (甘油) and fatty acids (脂肪酸).	1: enzyme action or products partly correct.	0: unused or incorrect; do not score by keywords.

A blank teacher score means pending, while an entered 0 means zero. Calculate the overall score only after reflection and required scoring are complete.

For teachers only. Do not give this part to students.

Scoring guide

Draft activity scoring guide for researcher review; it is not a validated SPSAI or CKT.

Item	Maximum	Scoring	Full credit	Partial credit	Zero / missing evidence
High temperature and enzyme denaturation	2	Teacher: post-learning reflection only	2: links the boiled-Y observations with protein denaturation and loss of function.	1: knows heat has an effect but does not explain how.	0: concept unused or wrong.
Revising the original explanation using data	2	Teacher: post-learning reflection only	2: addresses whether the original prediction is supported, makes specific comparisons and uses concepts to improve the explanation, with appearance limits.	1: comparison or revision is present but links are incomplete.	0: no useful revision or link to data.
SPS, new knowledge and overall totals	32	Totals	Six SPS areas = 24; four knowledge items = 8. Show overall only after reflection and all required scores are complete.	Blank means pending; 0 means zero. Teacher fields accept only whole numbers in range.	Invalid or non-whole numbers do not produce a valid total. Do not fill blanks with 0 just to complete scoring.
Colours, records and keeping teacher scores	0	Note	Excel uses green for full/correct, red for zero/incorrect, amber for partial; fills show categories.	Original, first and final answers and events are separate; digital time means visible activity time.	Teacher scores remain in the separately saved Excel file, and a new export does not include them. Student PDFs contain no scores.

A blank teacher score means pending, while an entered 0 means zero. Calculate the overall score only after reflection and required scoring are complete.